Nimra Faryal

AI Engineer | LLMs, Semantic Search, Vector Databases & Agentic AI
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Summary

Computer Science graduate from UET Taxila with a CGPA of 3.69 who builds LLM and agentic AI systems in Python. I have built a safety gateway that checks AI-agent tool calls, an LLM-powered interview system, and semantic search over a Qdrant vector database. I enjoy turning ideas into working prototypes.

Education

University of Engineering and Technology (UET) Taxila, Pakistan
BS Computer Science – CGPA 3.69/4.00 *Sep. 2022 – Jul. 2026*

- Merit Scholarship for **1st position in the CS-22 batch** (4th semester GPA 3.95).
- Grade A in NLP, Artificial Intelligence, Advanced Topics in CS, Big Data Analytics and Databases.

Technical Skills

LLMs: OpenAI API, Groq (Llama 3.3 70B), DeepSeek V3, prompt engineering, LLM scoring & evaluation, Whisper
Agentic AI: AI agent tool calls, guardrails, approval workflows, audit logging, FastAPI agent services
Semantic Search & Vector DBs: Qdrant, BERT embeddings, similarity search, RAG concepts
NLP & ML: Hugging Face Transformers, text classification, Scikit-Learn, PyTorch, model evaluation
Python & Data: Python, Pandas, NumPy, SQL, PostgreSQL, SQLite, data cleaning, Matplotlib
APIs & Web: FastAPI, Flask, REST APIs, JSON, SQLAlchemy, Pytest, Git/GitHub
Computer Vision: YOLOv8/v11 (detection, OBB, pose, segmentation), ByteTrack, Re-ID, OpenCV, Roboflow

Hackathons

AMD Developer Hackathon Act III – lablab.ai GitHub
ACTION GATE: AI Agent Security Gateway | Python, FastAPI, DeepSeek V3 *Oct. 2026*

- Built a **FastAPI** service that checks every **AI-agent tool call** before it runs: allow, ask for approval or block.
- Added a dry-run impact check, an **audit log** of every decision and **52 automated tests**.

AI Genesis Hackathon – lablab.ai
Multimodal Insurance Claims Processor | Python, Qdrant, BERT, OpenAI *Nov. 2025*

- Worked in a team on **semantic search** in the **Qdrant vector database**, finding similar past claims across text, images, audio and video to support fraud and routing decisions.

Projects

IoT Forest Monitoring Network (Final Year Project, team of 3) GitHub
Python, Flask REST API, PostgreSQL, Raspberry Pi

- **My role:** designed the **PostgreSQL database** and built the **Flask REST API** for sensor data.

AI Interview System (LLM Scoring & Ranking) GitHub
Python, Groq API (Llama 3.3 70B), Whisper, Streamlit

- Used an **LLM through an API** to write role-specific questions and **score each answer**.
- Turned speech into text with Whisper and exported results as **CSV and JSON reports**.

Sentiment Analysis Web App (NLP) GitHub
Python, Hugging Face Transformers, Streamlit

- Built a **text classifier** that labels input as positive or negative with a confidence score.

Walmart Weekly Sales Forecasting GitHub
Python, Pandas, Scikit-Learn, Matplotlib

- Cleaned and combined sales and economic data and forecast weekly sales with $\mathbf{R^2 = 0.945}$.

Computer Vision Projects

Farm Predator Detection LinkedIn Demo
Python, YOLO11, Roboflow, ByteTrack, Kaggle GPU
- Researched why a ready-made model failed, then **built and labelled my own dataset** and retrained it.

CCTV Suspicious Behaviour Detection LinkedIn Demo
Python, YOLO11-Pose, ByteTrack, OpenCV
- Scored risky actions from **pose keypoints** and zone rules, with stable tracking IDs and a live alert dashboard.

Drone-Based Vehicle Detection & Traffic Analytics LinkedIn Demo
Python, YOLO11s-OBB, OpenCV
- Detected vehicles in aerial drone footage with **rotated boxes** and turned detections into traffic counts.

AI Basketball Analytics & Live Commentary LinkedIn Demo
Python, YOLO, ByteTrack, Re-ID, Homography, ElevenLabs
- Tracked players through occlusions, measured speed and turned game events into **voiced commentary**.

Employee Safety (PPE) Detection LinkedIn Demo
Python, YOLOv11, PyTorch, OpenCV
- Trained a **custom YOLOv11 model on 7 classes** and flagged workers missing safety gear on site video.

Real-Time Traffic Analysis & Conveyor Orange Counting Demo (Traffic) | Demo (Oranges)
Python, YOLOv11, OpenCV, Object Tracking
- Counted vehicles with **unique ID tracking** and a counting line, and counted oranges on a conveyor belt.

AI Squat & Push-up Coach Demo (Squat) | Demo (Push-up)
Python, MediaPipe Pose, OpenCV
- Counted reps and **scored form out of 100** from pose keypoints, running live on a laptop CPU.

Certifications & Achievements

Certifications: Intro & Intermediate SQL (DataCamp); Introduction to Generative AI (Google Cloud)
Awards: 5th, Logic and Code Clash Round II (A&D Tech Innovations); PM Laptop Scheme Awardee
Leadership: Aspire Leaders Program Cohort 4 (2026); UET Women's Basketball Team (2023–25)